AIM-PRO
AI Literacy for Multidisciplinary Professional Readiness and Outreach

Task 5.1
Pedagogical requirements:

Training strategies for the development of AI competencies

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1. Introduction

As AI becomes a core part of everyday work and learning, support needs to move beyond teaching basic digital skills and **actively help individuals build strong AI competencies** (Markauskaite et al., 2022). This means preparing people not just to use AI tools, but to understand them, question them, and apply them responsibly (Fassbender, 2025). The question of how best to prepare people for working with AI remains open. Clearly, building AI competencies requires more than technical know-how: it demands a mix of **practical skills, critical thinking**, **ethical awareness,** and the **ability to adapt to fast-changing technologies** (cf. Howard et al., 2025). Support needs to combine hands-on experience with AI tools together with strategies that help individuals understand how these systems work, where they are useful, and where their limits and risks lie (Gandía, de Lucas Ancillo, & del Val Núñez, 2025).

Because every individual has a different background and comfort level with technology, there is no single approach that works for everyone. Earlier models for digital competence offered valuable guidance but given how quickly AI and digital technologies have evolved, digital strategies are now essential. This WP therefore aims to identify the most recent and effective approaches for developing AI-related competencies by revisiting and expanding previous research in this area. Specifically, we focus on a Synthesis of Qualitative Data (SQD); a model that brings together effective strategies to develop digital competencies (Tondeur et al., 2025).

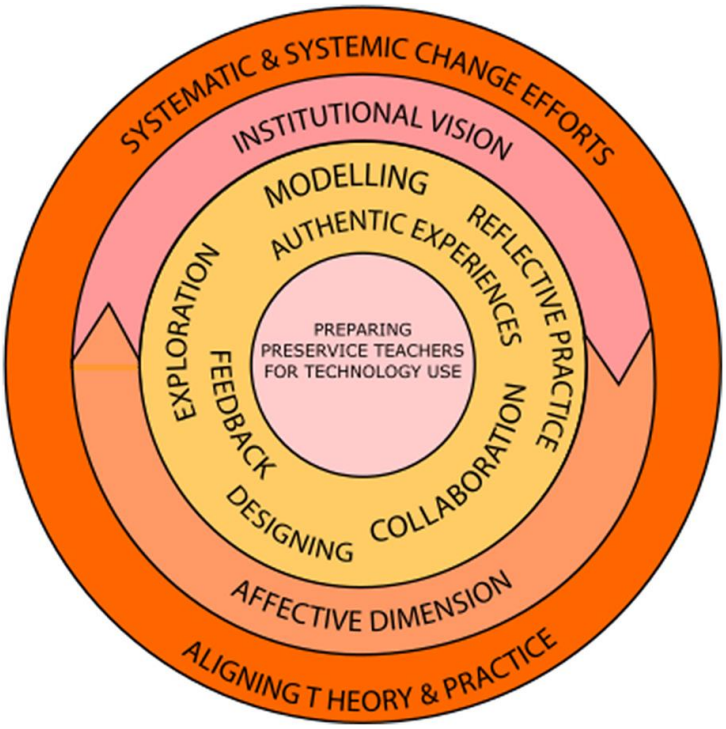


Fig. 1. SQD-model for digital technologies (Tondeur et al., 2025).

2. Training strategies to develop digital competencies

In the SQD2 model, the micro level focuses on the specific strategies used to prepare individuals to use digital technologies effectively into their practice (see Fig. 1). These strategies are categorized into five key syntheses that are applicable to developing Artificial Intelligence (AI) competencies:

(1) Familiarisation

Familiarisation is the initial stage where learners encounter digital tools through two main approaches: Modelling (observing experts use the tool) and Exploration (active, hands-on experimentation).

➔ Contextualised to AI: AI is explicitly identified as an "emerging technology" that requires this dual approach. Developing AI competency involves moving beyond just watching a demonstration; it requires hands-on experience with AI tools to

build confidence and understand their practical utility before applying them in professional settings.

(2) Reflective Practice

This strategy requires learners to critically reflect on the limitations and potential of technology while also considering their own "Digital Identity" as professionals.

➔ Contextualised to AI: A breakthrough in the SQD2 model is the use of AI as a scaffold for reflection. For example, learners can use AI-generated word clouds or AI-directed speech recognition to analyze their group discussions and lesson planning. This helps individuals develop a “digital agent” persona, where they use AI not just as a tool, but as a partner in analyzing their pedagogical choices.

(3) Collaborative Design

The model emphasizes that designing technology-rich materials should be a collaborative process to ensure the technology aligns with learning outcomes or other goals.

➔ Contextualised to AI: Developing AI competency involves collaborating with peers to design materials that integrate AI. This process requires pedagogical reasoning, understanding the “why” and “how” of using a specific AI tool to achieve a specific outcome, rather than using it for the sake of novelty.

(4) A Cycle of Authentic experiences

This strategy involves an iterative cycle of authentic experiences (using technology in real-world scenarios) followed by continuous feedback from mentors and peers.

➔ Contextualised to AI: To master AI, learners need opportunities for “real-world” application, such as online micro-teaching with AI assistance. The model suggests that human feedback is more valued than machine feedback in this process, helping the learner translate the experience into their practice.

(5) The Affective Dimension

The SQD2 model introduces the Affective Dimension as a core micro-level consideration, focusing on the Emotional Sphere (joy, stress, or frustration) and Attitudes/Beliefs of the learner.

➔ Contextualised to AI: AI integration often triggers strong emotional responses; positive experiences boost confidence, while negative ones can lead to frustration that hinders adoption. Furthermore, developing AI competency requires a "Sensitivity to Tech Gaps," acknowledging that disparities in digital competence and equity (e.g., access to advanced AI models) can marginalize certain students.

Below, we present AI-specific strategies grounded in SQD2, based on a qualitative synthesis of 40 A1 research papers (Zhou et al, in press).

3. Training strategies to develop AI competencies

The SQD2 model provides five strategies to prepare individuals to use digital technologies effectively into their practice. To prepare individuals to use AI in their practice iSQD (Zhou et al, in press) builds on these syntheses and extends them with the AI-specific strategies. Below these specific training strategies for AI-competencies we present concrete actions related to each of the strategies.

(6) Familiarisation

To familiarise individuals with AI technologies AI onboarding (building AI-knowledge and modelling practices), AI-supported scenarios (experiencing affordances and getting case-based instructions) and the progressive exploration (scaffolded and open) are needed.

➔ Systematic explanations of AI concepts, principles, application potential, and usage strategies and the observation of experts' or peers' AI practice generally form a first foundational step. While undertaking activities that map the tool's function with the individual's context and reviewing and analysing AI-specific

cases help discover scenarios for using AI. Finally, individuals are guided through structured support (e.g., worksheets) and encouraged to openly explore its use, based on their interests.

➔ Training strategy 1: "***progressively build AI knowledge and experiences, supported with targeted scenarios for using AI and structured guidance towards open exploration are essential to familiarise individuals with AI technologies"***

(7) Reflective Practice

To help professionals critically reflect on the potential and the limitations of technology, negotiations about AI's role, evaluations of its output and the documentation of the users' reflection are needed. So are participating in reflective discussions and AI-mediated reflections.

➔ AI not only helps individuals to analyse their (pedagogical) choices, but it also pushes them to reflect on their professional identity and on AI-generated (instructional) materials. Structured artefacts (e.g., reflective journals) help document reflections and group discussions provide multiple perspectives whereas reflective conversations with AI systems automate support and provide immediate feedback.

➔ Training strategy 2: ***"individuals must critically reflect on the potential, the limitations and place of AI for both their practice and their profession.***

(8) Collaborative Design

To ensure the technology aligns with the targeted outcomes, designing technology-rich materials should be a collaborative process between humans and supported by AI.

➔ AI competencies and skills linked to pedagogical reasoning are built through peer interactions and the sharing of AI experiences, instructional materials, prompting strategies, and design outcomes. While sustained learning communities provide with ongoing mutual support, resource sharing, and collective growth. Not only the interactions between humans provide opportunities to build skills, so does the interaction with AI: when co-creating

instructional content and making sure it aligns with the instructional goals, learner levels, cultural contexts, and curriculum standards. These iterative interactions are then again effective ways to develop prompting skills.

➔ Training strategy 3: ***"to design AI-rich materials, collaborative activities need to support interactions between humans and help them share resources and strategies, while utilizing the iterative interactions to design with the technology"***

(9) A Cycle of Authentic experiences

The use of AI in real-world scenarios (like AI-integrated teaching and evaluation) support individuals to responsibly integrate AI and enrich AI-content authentically.

➔ Practicing in low-risk environments, like microteaching, and systematically refining implementations in real settings help translate experiences into practice. Evaluating appropriateness, ethical implications, and limitations of AI use and enhancing AI-generated content with emotional tone and human sensitivity (to reduce mechanical or impersonal qualities) help with fitting AI-use into their authentic setting.

➔ Training strategy 4: "***Practicing and refining AI-integration for and adapting AI-content to their authentic setting feedback is needed to support individuals to use AI in their practice".***

(10) The Affective Dimension

Attending to effective responses remains a core micro-level consideration when using AI, focussing on emotional reactions, attitudes and beliefs, and self-efficacy.

➔ Emotions experienced by teachers during AI use, including anxiety, worry, enjoyment, satisfaction, and other emotions impact adoption. So do individuals' attitudes, perceptions, trust, and value beliefs toward AI tools and their confidence and self-assessment of their competencies to use AI tools in teaching.

➔ Training strategy 5: ***"emotional reactions, attitudes and beliefs, and self-efficacy should be attended to when preparing individuals for developing their AI competencies"***

In the following section we provide with concrete activities linked to each strategy.

4. List of activities to develop AI competencies based on iSQD training strategies

Strategy (synthesis)	Activity (category)	Activity (subcategory)	Description (implements iSQD strategies)
S1 Familiarization: ***progressively build AI knowledge and experiences, supported with targeted scenarios for using AI and structured guidance towards open exploration are essential to familiarise individuals with AI technologies***			
	C1 AI onboarding	SC1 Building AI knowledge	Systematic explanation of AI concepts, principles, application potential, and usage strategies.
		SC2 Modelling AI practices	Indirect AI usage experiences through modelling experts or peers using AI.
	C2 AI-supported scenarios	SC3 Experiencing affordances of AI tools	Presentation of AI tools and their functions to establish tool-function-context mappings.
		SC4 AI case-based instruction	Review and analysis of AI-integrated instructional cases demonstrating AI use in context.
S2 Reflective practice: ***individuals must critically reflect on the potential, the limitations and place of AI for both their practice and their profession.***			
	C3 AI progressive exploration	CS5 Scaffolded exploration	Guided AI task completion under structured support (e.g., worksheets, checklists, explicit task instructions).
		SC6 Open exploration	Opportunities for open exploration of AI tools based on individual interests, encouraging experimentation and discovery.
	C4 Reflective practice on AI	SC7 Role negotiation	Reflection on User-AI role boundaries and professional identity in AI-enhanced contexts.
		SC8 Output	Critical reflection on AI-generated

		evaluation	instructional materials.
		SC9 Documented reflection	Documentation and reflection on AI use through structured artefacts (e.g., reflective journals, portfolios, concept maps, blog entries).
		SC10 Reflective discussion	Group discussions foster critical reflection through multiple perspectives.
		SC11 AI-mediated reflection	Reflective conversation with AI systems for automated support and immediate feedback.
S3 Collaborative design: ***to design AI-rich materials, collaborative activities need to support interactions between humans and help them share resources and strategies, while utilizing the iterative interactions to design with the technology***			
	C5 Collaboration	SC12 Peer support	Building AI competencies and skills through peer interactions within learning.
		SC13 AI resource & strategy sharing	Sharing AI experiences, instructional materials, prompting strategies, and design outcomes.
		SC14 Community support	Establishment of sustained learning communities for ongoing mutual support, resource sharing, and collective growth.
	C6 AI-supported design	t	Co-creation of new instructional content with AI tools, emphasizing teacher-AI collaboration.
		SC16 AI-informed pedagogical adaptation	Adaptation and contextualization of AI-generated or existing materials to align with instructional goals, learner levels, cultural contexts, and curriculum standards.
		SC17 Iterative prompting	Iterative prompting and the refinement of prompts through multi-turn dialogue to develop effective AI interaction skills.

S4 A cycle of authentic experiences ***Practicing and refining AI-integration for and adapting AI-content to their authentic setting feedback is needed to support individuals to use AI in their practice***			
	C7 AI-integrated teaching cycle	SC18 AI-assisted microteaching	Microteaching activities involving AI-assisted instruction conducted in supportive and low-risk environments (e.g., workshops) to practice and refine approaches.
		SC19 AI-integrated classroom teaching	Implementation of AI-assisted instruction in authentic settings, involving sustained practice with real learners.
		SC20 Emotional enrichment of AI content	Enhancing AI-generated content and feedback with emotional tone and human sensitivity to reduce mechanical or impersonal qualities.
	C8 AI-integrated teaching improvement	SC21 AI-integrated lesson refinement	Systematic refinement of AI integration in instruction.
		SC22 Responsible AI use	Critical evaluation on the appropriateness, ethical implications, and limitations of AI use.
S5 Affective dimension ***emotional reactions, attitudes and beliefs, and self-efficacy should be attended to when preparing individuals for developing their AI competencies***			
	C9 Affective dimension	SC23 Emotional reactions	Emotions experienced during AI use, including anxiety, worry, enjoyment, satisfaction, and other emotions.
		SC24 Attitudes & beliefs	Attitudes, perceptions, trust, and value beliefs toward AI tools.
		SC25 Self-efficacy	Confidence and self-assessment of their competencies to use AI tools.
S6 Institutional vision (not included in the text yet)			
	C10 AI-	SC26 AI courses	AI courses with clearly defined

	integrated curriculum		weekly or modular arrangements and a progressive learning pathway.
		SC27 Educational frameworks	Systematic design of AI teaching courses based on established frameworks (e.g., IBL, TPACK).
		SC28 Discipline-specific AI integration	Design of differentiated AI training programs according to subject characteristics.
	C11 AI-focused training & incentives	SC29 AI-related professional learning	Provision of ongoing professional learning opportunities to keep pace with the rapid development of AI.
		SC30 Incentives	Institutional incentive policies (e.g., certification, rewards, or career development opportunities) encouraging to participate in AI training and instructional applications.
	C12 Institutional support	SC31 Institutional AI guidelines	Development of institutional policies and ethical frameworks for AI use, accompanied by normative guidance established through cross-departmental collaboration.
		SC32 Leadership support	Strategic support from institutional leadership, including resource provision and decision-making commitment, to promote the systematic implementation of AI integration.

5. Guidelines to implement iSQD strategies

(11) The iSQD strategies are interrelated

Several studies showed that helping individuals to use technology effectively isn't just about teaching one specific (SQD) strategy. Instead, the different parts of their training are strongly connected and work best when they reinforce each other. The Howard et al (2023) study shows this clearly: the categories are so intertwined that it's hard —to look at them separately. For example, "designing" often goes hand-in-hand

with “authentic experiences”. Overall, the message is that developing AI competencies doesn’t come from isolated digital SQD strategies, but from a coordinated set of strategies that help learners understand, practice, and apply AI applications.

(12) Adapting to Individual Profiles

The core of "one size does not fit all" is that learners have different profiles, unique combinations of attitudes, self-efficacy, and previous experiences (see Tondeur, Howard, & Yang, 2021). To illustrate, the Tondeur et al. (2012) study shows that:

- Learners with high attitudes towards technology tend to thrive through collaboration and peer-based (re)design of materials.
- Learners with low attitudes or lower self-efficacy towards technology often require an emphasis on continuous feedback and confirmation to mitigate feelings of insecurity and anxiety.

For learning to be truly inclusive, the model incorporates the Affective Dimension, which considers the emotional sphere of the learner (cf. Training strategy 5). It acknowledges that feelings such as joy, pride, stress, or frustration can substantially influence a person's ability to engage with technology. By fostering reflection on these emotional responses, training programs can help learners from all backgrounds overcome psychological barriers to adoption.

(13) Sensitivity to Equity

Sensitivity to equity means shifting focus from mere "access to devices" to a deeper sensitivity to technology gaps. This means explicitly addressing:

- Disparities in digital competence among learners.
- Intercultural awareness, ensuring that digital practices do not marginalize those from diverse cultural backgrounds.
- Support for students with learning difficulties, ensuring that technology acts as an enabler rather than a barrier.

(14) Moving Toward Adaptive, Smart Learning

The ultimate goal is to develop an adaptive model or "smart learning environment". This involves using data and technologies (like AI) to provide personalized, automated support. For example, a learner who is struggling might be prompted with specific feedback or a suggestion to collaborate, tailored to their individual needs in real-time.

(15)Program-Wide, Holistic Integration

To reach all learners, technology integration should not be a "stand-alone" event but must be woven systematically throughout the entire curriculum, institution, or company. This ensures that learners encounter digital strategies in various contexts, online, hybrid, or blended, which reinforces their understanding regardless of their preferred learning mode.

Below, we present how SQD2 and iSQD strategies can be linked to different profiles.

6. Persona's

Drawing on the training strategies presented above, here are three personas that illustrate the diverse needs of learners and how these strategies can be specifically applied to develop their **AI competencies**.

(16) Persona 1: Elena, the Sceptical HR Director

Profile: Elena is an HR Director in a traditional Small-to-Medium Enterprise (SME) with a heterogeneous workforce and varying levels of digital competence. While she sees potential value in AI for improving efficiency, she feels uncertain and anxious about fairness, accountability and employee trust. She hears colleagues expressing that the change may be too difficult. Her profile reflects low AI self-efficacy and a cautious attitude, driven more by responsibility concerns than resistance.

AI Competency Goal: To develop foundational AI literacy that enables her to integrate AI responsibly in HR practices, reduce anxiety (for herself and her colleagues) and make informed decisions about where AI can support human judgement.

Training strategies:

- ➔ **Familiarisation**: To build confidence without increasing anxiety, Elena benefits from HR-specific, guided AI scenarios (e.g. AI-assisted screening or reporting). Progressive familiarisation helps her understand AI affordances and constraints in a controlled context preventing cognitive overload. She needs to observe Role Models (like other successful SME managers) demonstrating tangible benefits of AI to build her confidence.
- ➔ **Reflective Practice:** Elena needs structured opportunities to critically reflect on AI-generated outputs and to negotiate HR–AI role boundaries. Guided reflection (e.g. short decision rationales or evaluation checklists) helps her make explicit where AI supports decision-making and where human responsibility must remain central.
- ➔ **Collaborative design:** Because Elena has a lower ICT attitude, the research indicates she will rely more heavily on feedback for confirmation and support as she develops integrated learning designs.
- ➔ **A Cycle of Authentic experiences:** Elena should pilot AI tools in low-stakes HR processes, followed by human feedback and ethical evaluation. This iterative refinement supports the translation of AI use into practice while safeguarding fairness, transparency, and employee trust.

- ➔ **Affective Dimension:** Training must focus on her "Emotional Sphere" to mitigate stress and frustration. Addressing her "Sensitivity to Tech Gaps" is also vital, as she needs to ensure that AI adoption doesn't marginalize employees with lower digital competence.
- ➔ Institutional vision

(17) Persona 2: **Marco**, the Proactive CS Student

Profile: Marco is an undergraduate Computer Science student who is highly confident with digital technologies and already uses AI tools in programming and problem-solving. He is tech-savvy, passionate about coding and motivated to innovate and differentiate himself. He faces increasing complexity in managing AI-supported workflows and maintaining oversight over AI-generated outputs. His profile reflects a high AI attitude and strong self-efficacy.

AI Competency Goal: To develop advanced AI competencies that enable him to effectively co-create with AI, critically evaluate AI-generated outputs, and maintain control over AI-supported work processes.

Training strategies:

- ➔ **Reflective Practice:** Marco should engage in critical evaluation of AI-generated outputs, focusing on accuracy, limitations, and appropriateness for specific tasks. AI-mediated reflection (e.g. reflective conversations with AI or documented self-reflection) supports him in analysing his design and problem-solving choices.
- ➔ **Collaborative design:** As a high-attitude learner, Marco will thrive through Collaborative Design tasks. Collaborative design activities with peers help Marco articulate design choices, discuss prompting strategies and refine AI-supported solutions. Sustained peer interaction helps deepen pedagogical and technical reasoning about AI use.
- ➔ **A Cycle of Authentic experiences:** Marco benefits from authentic AI-supported project work, including AI-assisted development and evaluation tasks. Iterative practice, combined with peer or mentor feedback, supports the refinement of AI integration in realistic contexts.

- ➔ **Affective Dimension** (Digital Identity)**:** His training should help him interpret his beliefs and values as he balances his human skills with AI automation.
- ➔ Institutional vision

(18) Persona 3: **Dr. Clara**, the Humanities Lecturer

Profile: Dr. Clara is a university lecturer in the humanities who is concerned about the influence of generative AI on student learning, assessment, and academic integrity. She is motivated to redesign her teaching but remains cautious about uncritical AI use. Her profile reflects strong pedagogical engagement, moderate AI confidence and the need to leverage AI for specialized research tasks.

AI Competency Goal: To develop AI competencies for responsible course and assessment design, enabling her to integrate AI thoughtfully while promoting critical AI literacy and maintaining disciplinary standards.

Training strategies:

- ➔ **Reflective Practice:** Dr. Clara requires structured opportunities for reflection on the role and limitations of AI in teaching and assessment. Documented reflection and group discussion support her in evaluating AI-generated materials and negotiating academic integrity concerns.
- ➔ **Collaborative design:** She needs to use frameworks like TPACK to align AI tools with her specific content knowledge and pedagogical goals. This ensures AI supports learning rather than replacing them. Collaborative design with colleagues supports this co-creation and adaptation of AI-integrated learning materials. Sharing experiences and strategies contributes to collective growth.
- ➔ **A Cycle of Authentic experiences:** She benefits from an iterative cycle where she experiments with AI-integrated teaching (e.g., in a flipped classroom) and receives Pedagogical Reasoning feedback (e.g., redesigned assignments, evaluating AI use in authentic settings) from a "collegial" network of mentors and peers.
- ➔ Institutional vision: At the meso level, she requires a clear Pedagogical Vision from her institution that provides ethical frameworks and structured professional learning.

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